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PanMETAI - a high performance tabular foundation model for accurate pancreatic cancer diagnosis via NMR metabolomics

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Dan-Ni Wu[1,2,3,4], Joey Jen[5], Erickson Fajiculay[5], Min-Fen Hsu[3], Ming-Chu Chang[6,7], Jen-Chen Yeh[5,8], Karen Sargsyan[5], Juozas Kupcinskas[9,10], Jurgita Skieceviciene[9,10], Ruta Steponaitiene[9], Egidijus Morkunas[9,10], Greta Gedgaudiene[9], Chao-Ping Hsu[5,11,12] ✉, Yu-Ting Chang[6,7,13] ✉ & Chun-Mei Hu[3] ✉

Late diagnosis and the lack of effective early detection techniques contribute to the poor prognosis of pancreatic ductal adenocarcinoma (PDAC). To address this challenge, we develop ^{1}H NMR-based metabolomics–AI platforms employing customized multilayer support vector machine (SVM), AutoGluon, and Tabular Foundation Model (TabPFN) frameworks. These platforms integrate serum metabolomic profiles—including small-molecule metabolites and lipoproteins—with clinical/biochemical parameters (age, CA19-9) and Activin A, derived from 902 participants (424 high-risk controls and 478 PDAC cases). Our TabPFN-based algorithm, PanMETAI, outperform state-of-the-art models. In the Taiwanese training and validation cohort, the model achieved an impressive AUC of 0.99 (95% CI: 0.98–0.99). Its robustness is further confirmed in a Lithuanian external validation cohort ($n = 322$), which yields an AUC of 0.93 (0.90-0.95). Notably, it identifies key signature patterns that improve early-stage (I/II) PDAC diagnosis and perform well with small sample sizes ($n = 50$). TabPFN-PanMETAI offers a rapid, accurate, and non-invasive tool for early PDAC detection, with strong potential for clinical application.

Pancreatic ductal adenocarcinoma (PDAC) remains a highly lethal malignancy with a dismal 13% 5-year survival rate[1], primarily due to late-stage diagnosis[2], underscoring a critical unmet need for reliable early detection as current diagnostic approaches lack sufficient specificity and sensitivity[3–5]. The serological biomarker, carbohydrate antigen 19-9 (CA19-9)[6,7], exhibits limited utility due to its elevated levels in various cancers and nonmalignant conditions[8,9], rendering it insufficient as a standalone diagnostic marker. Despite extensive studies have explored alternative PDAC biomarkers—including liquid biopsy approaches using protein and gene markers[10], cell-free DNA[11], microRNAs[12], and exosomes[13]—limitations in large-scale validation and clinical superiority over CA19-9 persist. As a result, current biomarkers

[1]Institute of Biochemical Sciences, National Taiwan University, Taipei, Taiwan. [2]Chemical Biology and Molecular Biophysics Program, Taiwan International Graduate Program, Academia Sinica, Taipei, Taiwan. [3]Genomics Research Center, Academia Sinica, Taipei, Taiwan. [4]Institute of Biological Chemistry, Academia Sinica, Taipei, Taiwan. [5]Institute of Chemistry, Academia Sinica, Taipei, Taiwan. [6]Department of Internal Medicine, College of Medicine, National Taiwan University, Taipei, Taiwan. [7]Department of Internal Medicine, National Taiwan University Hospital, Taipei, Taiwan. [8]Department of Chemistry, National Taiwan University, Taipei, Taiwan. [9]Institute for Digestive Research, Academy of Medicine, Lithuanian University of Health Sciences, Kaunas, Lithuania. [10]Department of Gastroenterology, Academy of Medicine, Lithuanian University of Health Sciences, Kaunas, Lithuania. [11]Division of Physics, National Center for Theoretical Sciences, Taipei, Taiwan. [12]Genome and Systems Biology Degree Program, Academia Sinica and National Taiwan University, Taipei, Taiwan. [13]National Taiwan University Hospital Hsin-Chu Branch, Hsinchu County, Taiwan. ✉e-mail: cherri@as.edu.tw; yutingchang@ntu.edu.tw; CMHU1220@as.edu.tw

remain inadequate for accurate, rapid, and cost-effective PDAC diagnosis in clinical settings.

Metabolomics, the study of metabolites and lipids within biological systems, offers a promising strategy for discovering PDAC biomarkers[14,15]. Disrupted metabolic homeostasis is increasingly recognized as a key contributor to PDAC tumorigenesis[16,17]. Several studies have identified metabolic and lipidomic biomarkers with potential diagnostic value[18–21]. However, reliance on a limited set of biomarkers is insufficient to capture the full complexity of PDAC and often suffers from issues of validation or reproducibility. This limits their effectiveness in distinguishing early-stage patients, particularly in high-risk populations, underscoring the need for a more comprehensive and robust diagnostic approach.

This study introduces a powerful approach to improve early detection of PDAC using high-resolution proton nuclear magnetic resonance (^{1}H NMR) spectroscopy. We generated comprehensive, label-free serum metabolomic fingerprints and integrated these with clinical/biochemical parameters and a prognostic protein biomarker to develop a multilayer diagnostic framework. This approach overcomes the limitations of conventional biomarker-driven methods by enabling practical interpretation of complex NMR spectra and integrating heterogeneous data to enhance early-stage cancer detection accuracy. We leveraged recent advancements in artificial intelligence (AI) to evaluate the diagnostic performance of several state-of-the-art models, including the transformer-based neural network (Tabular Foundation Model) TabPFN[22] and the AutoGluon AutoML platforms[23]. Through systematic optimization, we established a fast, effective, and highly accurate algorithm, PanMETAI (a TabPFN-based algorithm integrating NMR metabolomics, age, CA19-9, and Activin A) for PDAC detection.

Results

Developing machine learning approaches for enhanced pancreatic cancer detection using metabolic features, Age, CA19-9, and Activin A

An overview of the study design is presented in Fig. 1. Serum samples from 350 participants in the exploration cohort (Table 1) and 552 participants in the validation cohorts (Table 2) were divided into three datasets for machine learning (ML) development and validation (Fig. 1A). To simulate a clinically challenging diagnostic setting, we adopted a structured data-splitting strategy to train a generalized ML model. The first cohort was divided into a training dataset [$n = 200$; 100 high-risk controls (HRC) and 100 patients with stage I/II PDAC] and a developing (validation) test dataset ($n = 150$; 50 HRC and 100 patients with stage III/IV PDAC). The second cohort was used as an independent, blinded validation dataset ($n = 552$), designed to evaluate the model's performance in real-world clinical scenarios. Each dataset included both HRC (non-cancer) and PDAC (cancer) samples. Samples with missing values in clinical/biochemical parameters were excluded from ML prediction analyses.

For machine learning input, we collected three data modalities: NMR-based metabolomic spectra, protein biomarker levels, and clinical/biochemical parameters. Metabolomic profiling using 600 MHz NMR with a high-throughput SampleJet system generated both ^{1}H NMR-M (small molecule metabolites) and NMR-M + L (metabolites plus lipoproteins) spectra. These raw spectra were then processed through a standardized pipeline involving internal standard (TSP-d4) normalization, signal trimming, and water signal removal, ultimately yielding 71,074 (Supplemental Fig. 1) and 42,500 (Supplemental Fig. 2) features, respectively. Given Activin A's reported association with early-stage PDAC and poor prognosis[24,25], suggesting it plays a critical role in pancreatic cancer development. Therefore, serum Activin A level was included as a protein feature (Supplemental Fig. 3). In addition, recognizing eight clinical/biochemical parameters (age, gender, smoking, Body Mass Index (BMI), CA19-9, Carcinoembryonic antigen (CEA), Hemoglobin A1c (HbA1c), and fasting glucose (GluAC)) monitored by physicians may correlate with PDAC progression in high-risk individuals, so we evaluated these factors. This analysis, using Silhouette scores on UMAP clustering, identified age and CA19-9 as the variables offering the best discrimination between cancer and non-cancer cases (score: 0.35, Supplementary Fig. 4). Therefore, our final integrated dataset for model development comprised NMR-M, NMR-M + L, Activin A, age, and CA19-9 (Fig. 1A).

To handle the high dimensionality and heterogeneity of our input data, particularly the extensive spectral features (~40K-70K), we developed a multilayer support vector machine (SVM) framework (Fig. 1B, Materials and Methods). This architecture employed tailored feature selection strategies based on spectral type: SelectKBest for NMR-M and recursive feature elimination (RFE) for NMR-M + L (Supplemental Fig. 5). The resulting selected features were transformed into decision values by individual SVMs and subsequently integrated with clinical/biochemical parameters and protein features. This design aimed to prevent any single data modality from disproportionately influencing the model while retaining comprehensive spectral information, thus ensuring a balanced contribution to the prediction. The multilayer structure also enabled independent optimization and hyperparameter tuning of each SVM component before fusion, enhancing model development and flexibility.

To benchmark the performance of our multilayer SVM framework, we compared it against several established machine learning algorithms, including k-nearest neighbors (kNN), gradient boosting (GB), decision trees, random forests, and neural networks. Most of these were implemented using the AutoGluon platform, which provides robust and user-friendly ML workflows. AutoGluon's ensemble model (14 models) utilized the same feature selection approach as our multilayer architecture (Supplementary Fig. 5). Finally, we also evaluated TabPFN, a transformer-based method incorporating attention layers and in-context learning. Due to TabPFN's input size limitation (<500 features), we applied a modified feature selection strategy (Supplemental Fig. 6), selecting 215 features from NMR-M and 255 from NMR-M + L for integration with the clinical/biochemical parameters and protein data (Fig. 1B).

TabPFN demonstrates superior performance in PDAC detection

We evaluated the performance of various machine learning algorithms on the Developmental Test (Dev-Test) dataset. Our multilayer SVM demonstrated robust predictive capabilities, achieving an AUC of 0.96 (95% CI: 0.93–0.99), sensitivity of 0.96 (0.92–0.99), specificity of 0.83 (0.71–0.94), an overall accuracy of 0.92 (0.88–0.96), and a J-Statistic of 0.79 (0.67–0.90). Among the AutoGluon 1.1.0 models, XGBoost exhibited the highest performance, with an AUC of 0.97 (0.93–0.99) and a J-Statistic of 0.83 (0.73–0.93). TabPFN outperformed all other tested methods across nearly all metrics (Supplemental Table 1). It demonstrated high performance, achieving an AUC of 0.98 (0.94–1.00), sensitivity of 0.98 (0.95–1.00), specificity of 0.90 (0.81–0.98), accuracy of 0.96 (0.92–0.99), and a J-Statistic of 0.88 (0.78–0.96) when integrating all data modalities, while notably requiring less training time and no hyperparameter tuning.

To validate the top three Dev-Test algorithms (multilayer SVM, XGBoost, and TabPFN), we conducted a blind test and confusion matrix analysis (Fig. 2 and Table 3), which confirmed TabPFN's superior performance. In this test, TabPFN achieved an high AUC of 0.99 (0.98–0.99), accuracy of 0.94 (0.91–0.95), sensitivity of 0.93 (0.89–0.96), and specificity of 0.94 (0.91–0.97). In contrast, the multilayer SVM and XGBoost methods were less effective, reaching overall accuracies of approximately 0.90 and 0.93, respectively. The final algorithm, named TabPFN-PanMETAI (integrating NMR metabolomics, age, CA19-9, and Activin A), demonstrated significant robustness and generalizability in the external validation test on a Lithuanian cohort (Table 2), yielding an AUC of 0.93 (0.90–0.95), sensitivity 0.90 (0.85–0.94), and specificity 0.83 (0.77–0.89), with consistent

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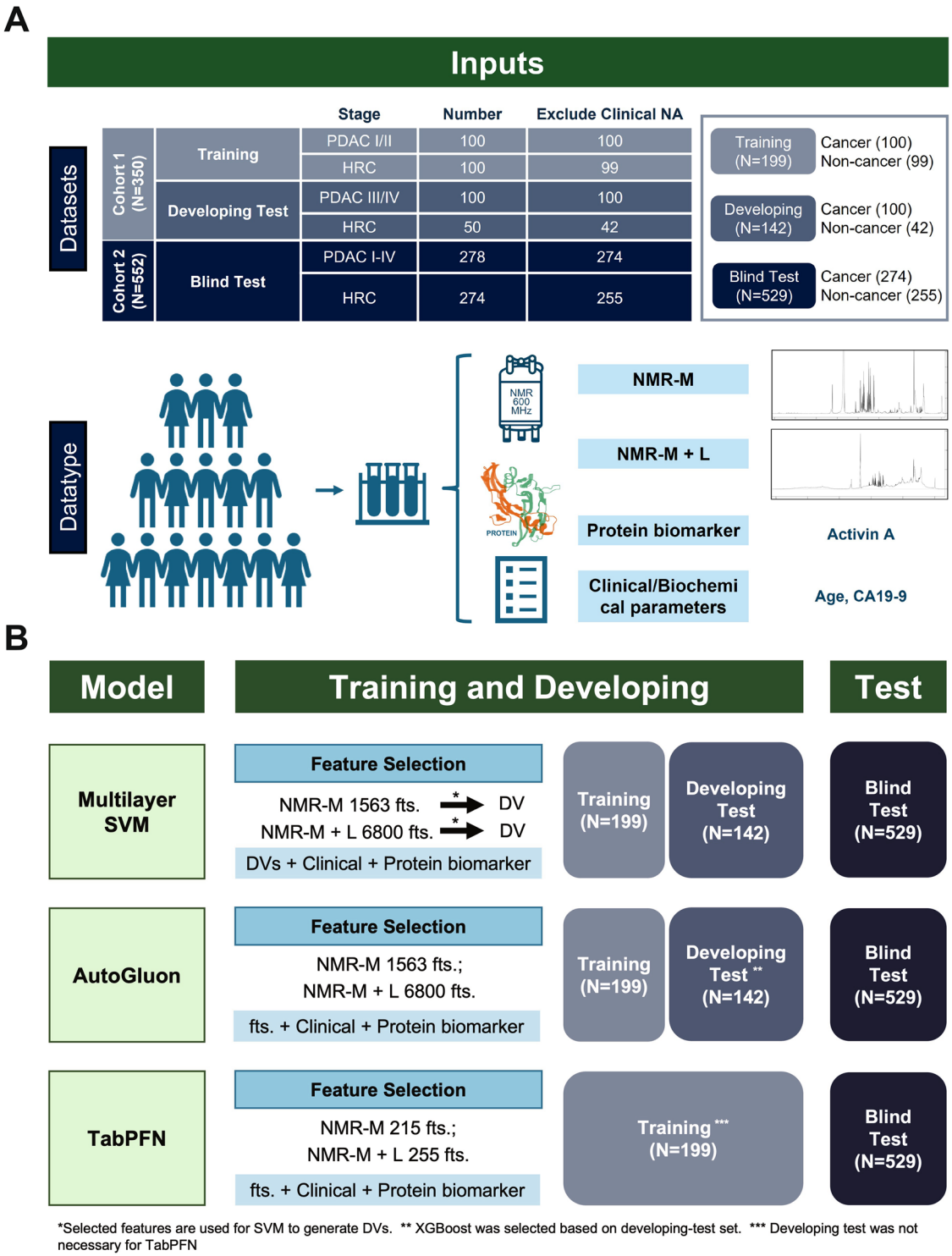

Fig. 1 | Schematic of study design. A Data types include NMR-L, NMR-M + L, Protein data (Activin A), and clinical/biochemical parameters (CA19-9 and age). Datasets include training, developing test, and blind test datasets for model development. **B** This diagram shows different models' development, including feature selection and dataset separation. All three models were trained on the Training set. The Dev-Test set served model tuning (except TabPFN). Decision Value (DV) indicates a sample's distance from the SVM separating hyperplane. "ft." represents the number of features used in each corresponding model. NMR denotes to nuclear magnetic resonance. NMR-M represents the ^{1}H NMR spectra of small molecule metabolites. NMR-M + L represents the ^{1}H NMR spectra of small molecule metabolites and lipoproteins. HRC indicates high risk control. PDAC I/II/III/IV indicates pancreatic ductal adenocarcinoma patients at stages I to IV.

performance even for early-stage (Stage I/II) patients (Table 4). The consistent success across two geographically and behaviorally distinct populations, Taiwan and Lithuania, provides compelling evidence of TabPFN-PanMETAI's strong potential for widespread clinical implementation.

^{1}H NMR metabolomics in TabPFN-PanMETAI enhances early-stage PDAC detection

TabPFN-PanMETAI demonstrated high performance in the Taiwanese cohort across different PDAC stages. For detecting late-stage PDAC (III/IV), the model shown high effectiveness, achieving an

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Table 1 | Characteristic clinical demographic data for pancreatic ductal adenocarcinoma (PDAC) patients and high-risk controls (HRC) in the 1st exploration cohort

	1st Exploration cohort (*N* = 350)				
	HRC (*n* = 150)	PDAC stage I-II patients (*n* = 100)	PDAC stage III-IV patients (*n* = 100)	*P*[a] (PDAC I-II vs. HRC)	*P*[a] (PDAC III-IV vs. HRC)
Sex (M/F)-no. (%)	60 (40%)/ 90 (60%)	58 (58%)/ 42 (42%)	59 (59%)/ 41 (41%)	0.005**	0.003**
Age, mean (SD), year	47.57 (13.53)	62.71 (11.42)	62.93 (12.54)	< 0.0001****	< 0.0001****
BMI, mean (SD), kg/m^2	23.05 (3.81)	22.31 (4.09)	21.93 (3.49)	0.154	0.021*
Smoke (0/1)-no. (%)	133 (89%)/ 17 (11%)	66 (66%)/ 34 (34%)	64 (64%)/ 35 (35%)	< 0.0001****	< 0.0001****
CA19-9, mean (SD), U/mL	34.37 (161.42)	1214.13 (3238.71)	4682.29 (7237.12)	< 0.0001****	< 0.0001****
HbA1c, mean (SD), %	5.65 (0.84)	6.86 (1.41)	6.73 (1.47)	<0.0001****	< 0.0001****
CEA, mean (SD), ng/mL	2.49 (1.86)	11.38 (55.42)	45.56 (175.25)	0.058	0.004**
Fasting glucose, mean (SD), mg/dL	97.17 (22.35)	133.87 (57.16)	122.88 (39.95)	< 0.0001****	< 0.0001****
Activin A, mean (SD), pg/mL	214.79 (18.05)	499.69 (720.39)	652.70 (646.50)	< 0.0001****	< 0.0001****
OS, median (SD), months[b]	--	18.54 (13.12)	5.57 (11.43)	--	--

[a]*P* < 0·05*, *P* < 0·01**, *P* < 0·0001**** The statistical differences were calculated using two-tailed Student's T-test. [b]*P* < 0·0001**** between stage I-II PDAC patients and stage III-IV PDAC patients. *HRC* High-Risk Controls, *PDAC* Pancreatic Ductal Adenocarcinoma, *M* Male, *F* Female, *BMI* Body Mass Index, *CA19-9* Carbohydrate Antigen 19-9, *HbA1c* Glycated Hemoglobin, *CEA* Carcinoembryonic Antigen, *OS* Overall Survival.

Table 2 | Characteristic clinical demographic data for pancreatic ductal adenocarcinoma (PDAC) patients and high-risk controls (HRC) in the 2nd validation and 3rd external validation cohorts

	2nd Validation cohort (*N* = 552)				
	HRC (*n* = 274)	PDAC stage I-II patients (*n* = 37)	PDAC stage III-IV patients (*n* = 241)	*P*[a] (PDAC I-II vs. HRC)	*P*[a] (PDAC III-IV vs. HRC)
Sex (M/F)-no. (%)	116 (42%)/ 158 (58%)	19 (51%)/ 18 (49%)	134 (56%)/ 107 (44%)	0.301	0.003**
Age, mean (SD), year	48.02 (14.86)	62.14 (10.77)	64.53 (12.73)	< 0.0001****	< 0.0001****
BMI, mean (SD), kg/m^2	23.20 (4.21)	23.17 (3.64)	22.40 (3.67)	0.965	0.028*
Smoke (0/1)-no. (%)	220 (80%)/ 47 (17%)	24 (65%)/ 13 (35%)	180 (75%)/ 61 (25%)	0.012*	0.034*
CA19-9, mean (SD), U/mL	204.21 (1473.95)	788.62 (2138.61)	5202.36 (8541.21)	0.036*	<0.0001****
HbA1c, mean (SD), %	5.7 (0.9)	6.9 (1.5)	6.8 (1.8)	< 0.0001****	< 0.0001****
CEA, mean (SD), ng/mL	7.76 (77.54)	3.92 (3.05)	78.90 (419.51)	0.764	0.008**
Fasting glucose, mean (SD), mg/dL	97.0 (23.6)	123.5 (41.1)	133.6 (67.8)	< 0.0001****	< 0.0001****
Activin A, mean (SD), pg/mL	240.04 (103.90)	398.66 (284.45)	991.83 (975.13)	< 0.0001****	< 0.0001****
OS, median (SD), months[b]	--	19.40 (15.46)	4.77 (11.84)	--	--
	3rd External validation cohort from Lithuania (*N* = 322)				
	HRC (*n* = 150)	PDAC stage I-II patients (*n* = 62)	PDAC stage III-IV patients (*n* = 110)	*P*[a] (PDAC I-II vs. HRC)	*P*[a] (PDAC III-IV vs. HRC)
Sex (M/F)-no. (%)	75 (50%)/ 75 (50%)	32 (52%)/ 30 (48%)	51 (46%)/ 59 (54%)	0.832	0.564
Age, mean (SD), year	67.07 (7.95)	66.77 (9.88)	67.53 (11.27)	0.818	0.705
CA19-9, mean (SD), U/mL	18.57 (78.55)	62187.17 (110624.04)	46051.39 (101380.91)	<0.0001****	<0.0001****
Activin A, mean (SD), pg/mL	334.70 (214.39)	663.53 (599.01)	874.55 (1333.18)	<0.0001****	<0.0001****

[a]*P* < 0·05*, *P* < 0·01**, *P* < 0·0001**** The statistical differences were calculated using two-tailed Student's T-test. [b]*P* < 0·0001**** between stage I-II PDAC patients and stage III-IV PDAC patients.

accuracy of 0.95 (0.93–0.97), a sensitivity of 0.95 (0.92–0.97), a specificity of 0.95 (0.92–0.97), and an AUC of 0.99 (0.98–0.99). Importantly, TabPFN-PanMETAI also maintained high diagnostic efficacy for early-stage PDAC (I/II), achieving an accuracy of 0.92 (0.89–0.95), sensitivity of 0.78 (0.64–0.91), specificity of 0.95 (0.92–0.97), and an AUC of 0.95 (0.90–0.98) (Supplemental Table 2). Comparable results were also observed in the Lithuania cohort (Table 4). To evaluate which data modalities drove the model's superior performance, we analyzed early and late PDAC detection using NMR metabolomics alone, clinical/biochemical parameters and protein data alone, and the fully integrated model. While our findings revealed that clinical/biochemical parameters and protein data alone were sufficient for achieving high performance in detecting late-stage (III/IV) PDAC, the integration of all data types led to a significant increase in detection sensitivity specifically for early-stage PDAC (Fig. 3 and Supplemental Table 2). This crucial result strongly supports the essential role of capturing subtle metabolic alterations within the NMR spectra, confirming that the

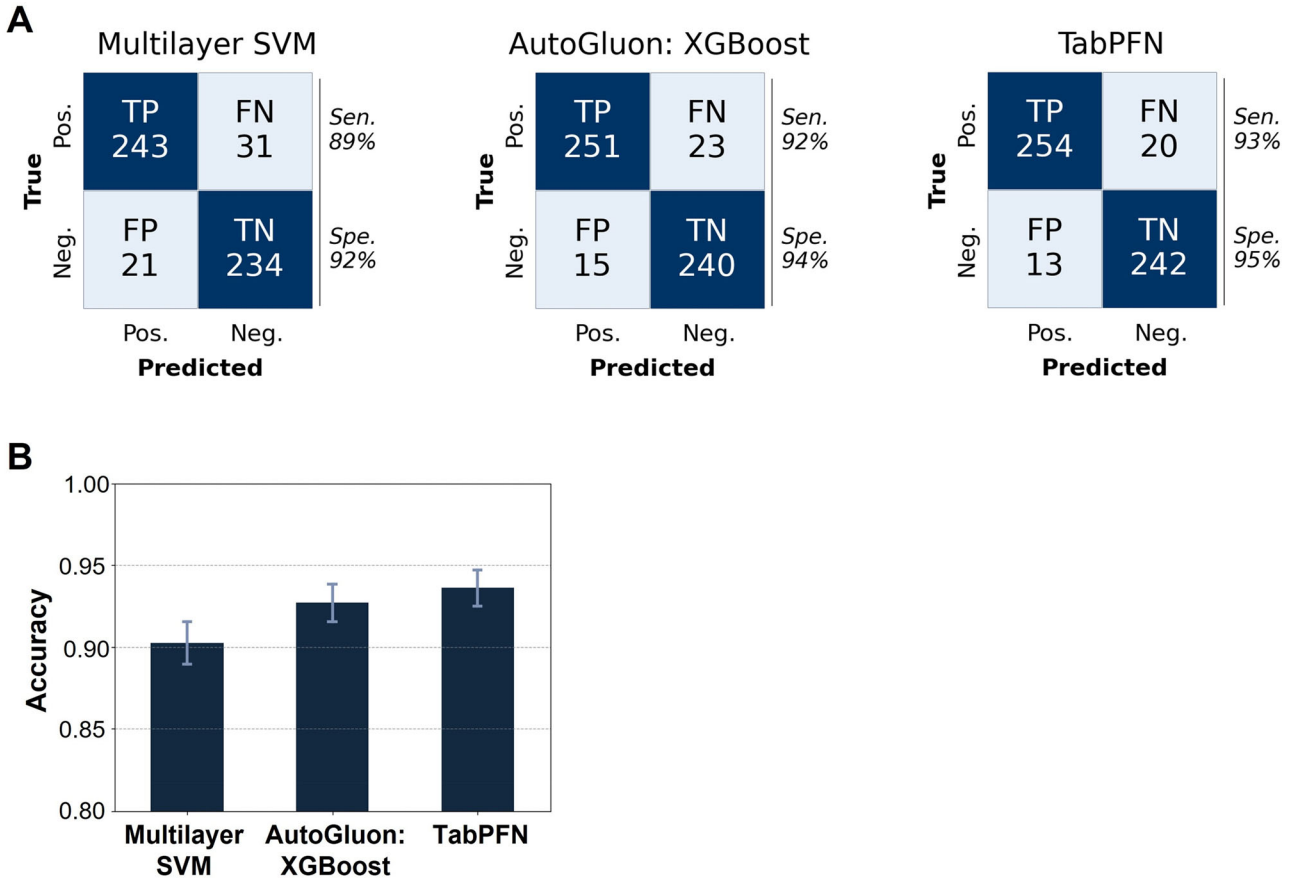

Fig. 2 | Diagnostic performance of the three outperformed models on the blind test dataset. A Confusion matrices for the three outperformed models evaluated on the independent blind test dataset. Pos. positive, Neg. negative, Sen. sensitivity, Spe. specificity. **B** Predictive accuracy comparison of the three outperformed models on the blind test dataset. The class ratio (cancer and noncancer groups) within the blind test dataset is balanced, the performance metric was presented as accuracy. Given that the blind test datasets are not comparable in structure or composition, statistical analyses are not applied in this context. Accuracy is presented as mean and standard deviation (SD) derived from 1000 bootstrap replicates.

Table 3 | Performance and confidence intervals of three models on the blind test dataset

Model	Accuracy	Sensitivity	Specificity	AUC	F1 score
Multilayer SVM	0.90 (0.88–0.93)	0.89 (0.85–0.92)	0.92 (0.88–0.95)	0.95 (0.94–0.97)	0.90 (0.88–0.93)
AutoGluon: XGboost	0.93 (0.90–0.95)	0.92 (0.88–0.95)	0.94 (0.91–0.97)	0.98 (0.97–0.99)	0.93 (0.90–0.95)
TabPFN	0.94 (0.91–0.95)	0.93 (0.89–0.96)	0.94 (0.91–0.97)	0.99 (0.98–0.99)	0.94 (0.91–0.96)

Data represents model performance, with error bars indicating 95% confidence intervals derived from 1000 bootstrap resampling.

Table 4 | Diagnostic performance of TabPFN-PanMETAI in the external Lithuanian cohort (*N* = 322)

PDAC Stages	Accuracy	Sensitivity	Specificity	AUC	F1 score
I/II/III/IV	0.87 (0.83–0.90)	0.90 (0.85–0.94)	0.83 (0.77–0.89)	0.93 (0.90–0.95)	0.88 (0.84–0.91)
I/II	0.83 (0.78–0.88)	0.84 (0.74–0.93)	0.83 (0.78–0.89)	0.91 (0.86–0.95)	0.75 (0.66–0.83)
III/IV	0.87 (0.83–0.91)	0.93 (0.88–0.97)	0.83 (0.77–0.89)	0.93 (0.89–0.97)	0.86 (0.81–0.90)

Data represents model performance, with error bars indicating 95% confidence intervals derived from 1000 bootstrap resampling.

inclusion of NMR data in TabPFN-PanMETAI is fundamental to enhancing early diagnosis.

Metabolomic Feature Selection and SHAP Analysis uncover TabPFN-PanMETAI's Biological Insights

Given the crucial role of NMR metabolomics within TabPFN-PanMETAI for effective cancer classification, we investigated the specific features prioritized by this algorithm to gain insights into its decision-making process and enhance interpretability. To understand why NMR-based metabolomic data strengthens the model's early predictive capabilities, we mapped the 255 and 215 metabolic features selected by TabPFN onto the ^{1}H NMR-M + L and NMR-M spectra, respectively (Fig. 4A). The corresponding metabolite identifications are detailed in the Fig. 4 legend, revealing a total of 11

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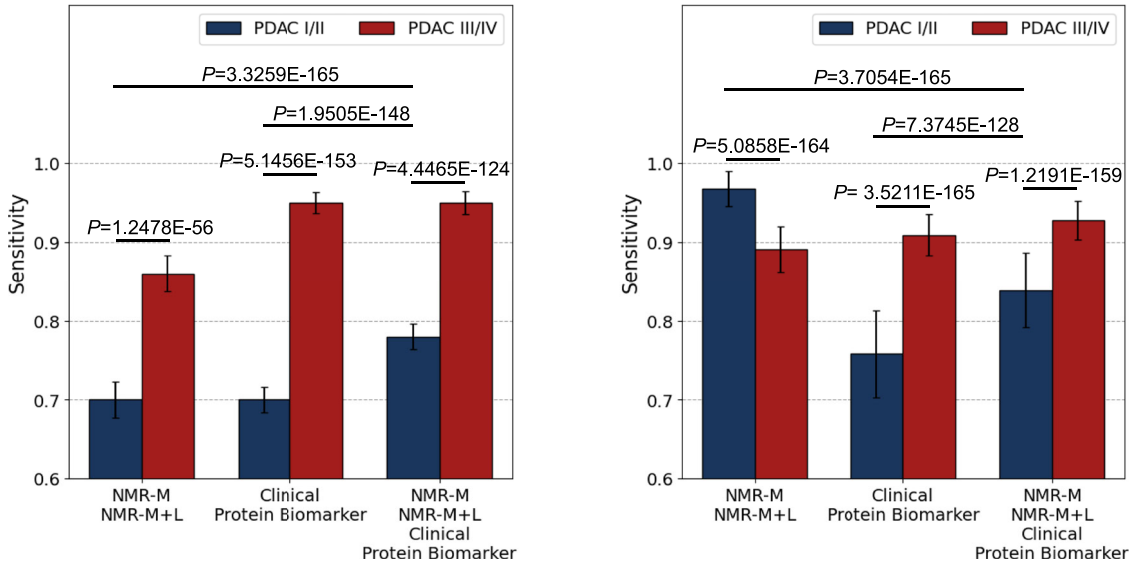

Fig. 3 | ^{1}H NMR metabolomic data improves early-stage PDAC detection by TabPFN-PanMETAI model. Comparison of the sensitivity of the TabPFN-PanMETAI in early versus late-stage detection across different datatype inputs. Sensitivity is evaluated with NMR data alone, combining clinical/biochemical parameters and protein biomarker data, and all three data types. Accuracy is presented as mean and standard deviation (SD) derived from 1000 bootstrap replicates. The *P* value was calculated using a two-tailed Wilcoxon Signed Rank test. TabPFN-PanMETAI, a TabPFN-based algorithm.

metabolites and 2 lipids from NMR-M + L, and 6 metabolites from NMR-M.

Subsequent quantification of these identified metabolites using the Bruker IVDr platform, cross-referencing with the IVDr database, revealed that 10 were quantifiable and exhibited statistically significant differences between cancer and high-risk control (HRC) groups (Fig. 4B). Notably, these metabolic alterations proved stage-dependent and align with prior research[26–31]. In early-stage (Stages I/II) PDAC, distinct changes were observed, including significant decreases in HDL and glutamine levels and marked elevation of lactic acid, TMAO, ornithine, glutamic acid, and glucose. In contrast, the metabolic profile became dynamic in advanced stages: some metabolites (like TMAO and Threonine) showed a trend toward normalization, while others (such as glutamic acid and glucose) remained persistently elevated. This finding highlights the dynamic role of metabolism throughout PDAC progression, emphasizing its crucial involvement in the early stages of tumor development, a pattern further confirmed by similar metabolite changes observed in the Lithuania cohort (Supplementary Fig. 7).

To further elucidate the TabPFN-PanMETAI model's decision-making process, we performed SHAP (SHapley Additive exPlanations) analysis[32] to assess the contribution of individual features. As depicted in Fig. 4C and Supplemental Table 3, the SHAP analysis highlighted those features at NMR-M + L ppm 1.23/3.27/3.74 (corresponding to HDL, VLDL, Glucose, TMAO, and Alanine) and NMR-M ppm 2.93/2.96 (corresponding to L-Asparagine, 3-Aminoisobutanoic acid, 2-Oxoglutaric acid, and L-Tyrosine) exerted a substantial impact on the model's ability to distinguish PDAC from high-risk populations. These findings underscore the interpretability of the TabPFN-PanMETAI model and the biological relevance of its feature selection, bolstering its translational potential in precision medicine.

To gain deeper biological insights, we conducted metabolic pathway and interaction network analyses using MetaboAnalyst 6.0 and Ingenuity Pathway Analysis (IPA), as illustrated in Fig. 5. The metabolic pathway enrichment map (Fig. 5A) confirmed that the SHAP-annotated metabolites were significantly enriched in key dysregulated pathways in PDAC, including the Glucose-Alanine Cycle, Phenylalanine and Tyrosine Metabolism, Aspartate Metabolism, and the Malate-Aspartate Shuttle, thus reinforcing the relevance of global alterations in amino acids and energy pathways[33,34]. The metabolic network (Fig. 5B) further highlighted molecules such as L-glutamate, alanine, pyruvate, cholesterol, and triglycerides as central hubs of dysregulation within mitochondrial and lipid metabolism. Finally, Fig. 5C summarized the disease and biofunctions derived from the SHAP features, with top enriched categories including cancer, organismal injury, and lipid metabolism, consistent with hallmark biological themes in PDAC progression. Collectively, these multi-layered analyses support the mechanistic relevance of TabPFN-PanMETAI's feature selection and provide crucial system-level insights into PDAC metabolic dysregulation.

Robust TabPFN performance with small datasets

Developing robust diagnostic models for low-incidence cancers like PDAC often faces the challenge of limited patient cohorts at individual centers. Traditional machine learning methods typically require large datasets for effective generalization, a constraint difficult to overcome in single-center studies. In contrast, TabPFN[22] has demonstrated competitive performance on small tabular datasets without hyper-parameter tuning, utilizing in-context learning from synthetic priors to approximate Bayesian inference. To assess the impact of training set size on our TabPFN-PanMETAI model's early PDAC detection capabilities, we conducted a subsampling analysis. We randomly selected varying numbers of stage I/II PDAC patients ($n = 3$–199) for training and evaluated the model on an independent blind test set over 20 repetitions. Our analysis revealed a rapid improvement in model performance (AUC, sensitivity, specificity, accuracy) with increasing training sample size. Notably, TabPFN-PanMETAI achieved stable high accuracy (≈0.90) with as few as 50–70 training cases (Fig. 6, Supplemental Fig. 8). The consistent and robust performance observed with such a limited training dataset ($n = 50$) is remarkable. These results underscore the potential of TabPFN's in-context learning paradigm for enabling robust diagnostic modeling in small clinical cohorts, offering a solution for medical applications where large datasets are not feasible.

Discussion

Metabolic reprogramming stands as a fundamental hallmark of cancer, reflecting its remarkable adaptive capacity to thrive within altered

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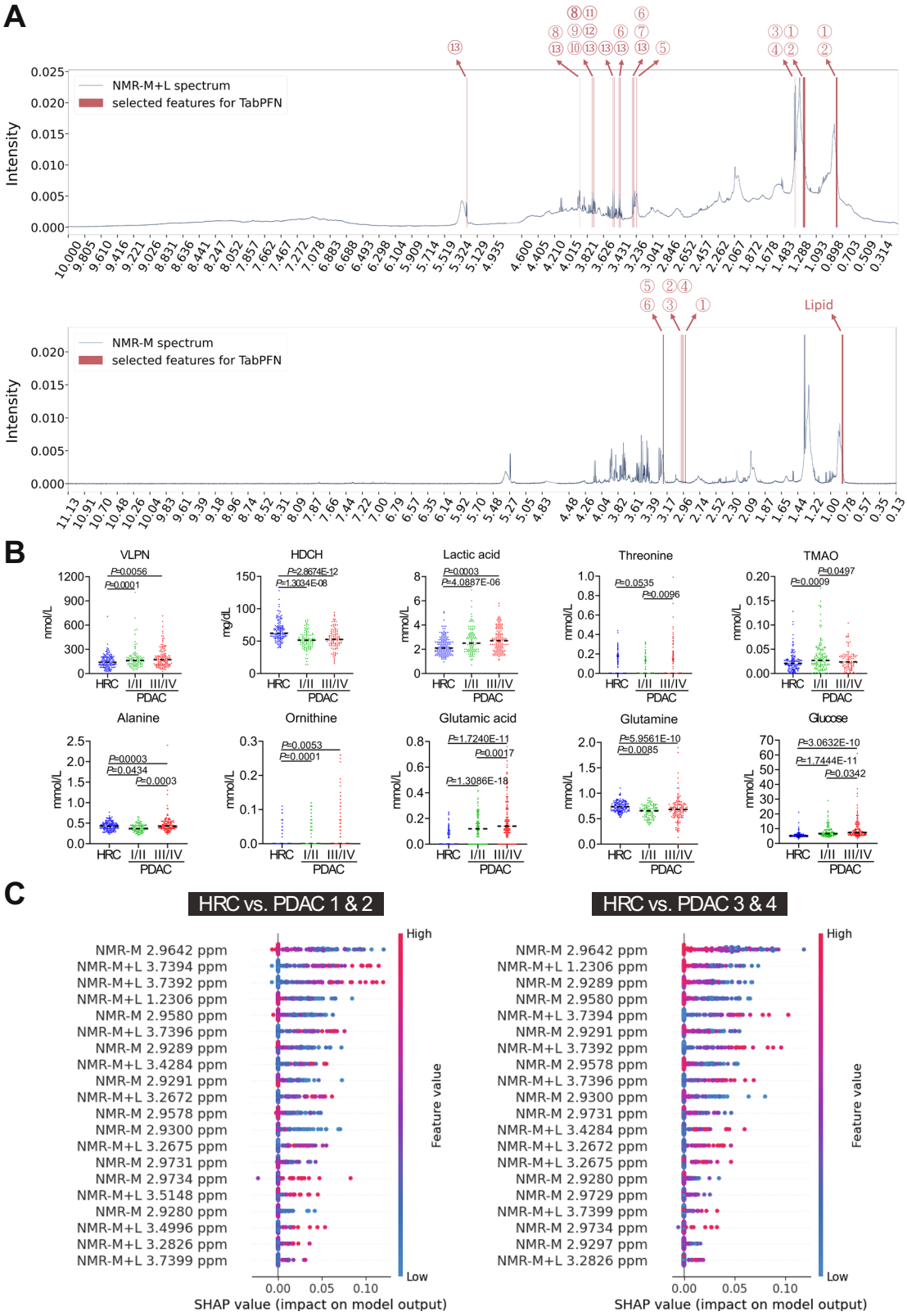

Fig. 4 | TabPFN-PanMETAI identified key metabolic features in ^{1}H NMR for PDAC detection. A The TabPFN-PanMETAI model selected the following metabolic features from serum NMR-M + L spectrum: ① Very low-density lipoprotein (VLDL), ② High-density lipoprotein (HDL), ③ Lactic acid, ④ Threonine, ⑤ Glycerophosphocholine (GPC), ⑥ Taurine, ⑦ Trimethylamine N-oxide (TMAO), ⑧ Glycogen, ⑨ Alanine, ⑩ Ornithine, ⑪ Glutamic acid, ⑫ Glutamine, and ⑬ Glucose. From NMR-M spectrum, the selected metabolic features were: ① L-Asparagine, ② 3-Aminoisobutyric acid, ③ 2-Oxoglutaric acid, ④ L-Tyrosine, ⑤ Choline, and ⑥ Phosphorylcholine. **B** Serum metabolites showing significant alterations were quantified using the Bruker IVDr platform, and statistical analysis was performed using a two-tailed Student's t-test with the exact *P*-value presented. **C** SHAP (SHapley Additive exPlanations) analysis highlights the impact of individual metabolic features in the NMR spectrum on TabPFN-PanMETAI's predictive accuracy for early- and late-stage PDAC.

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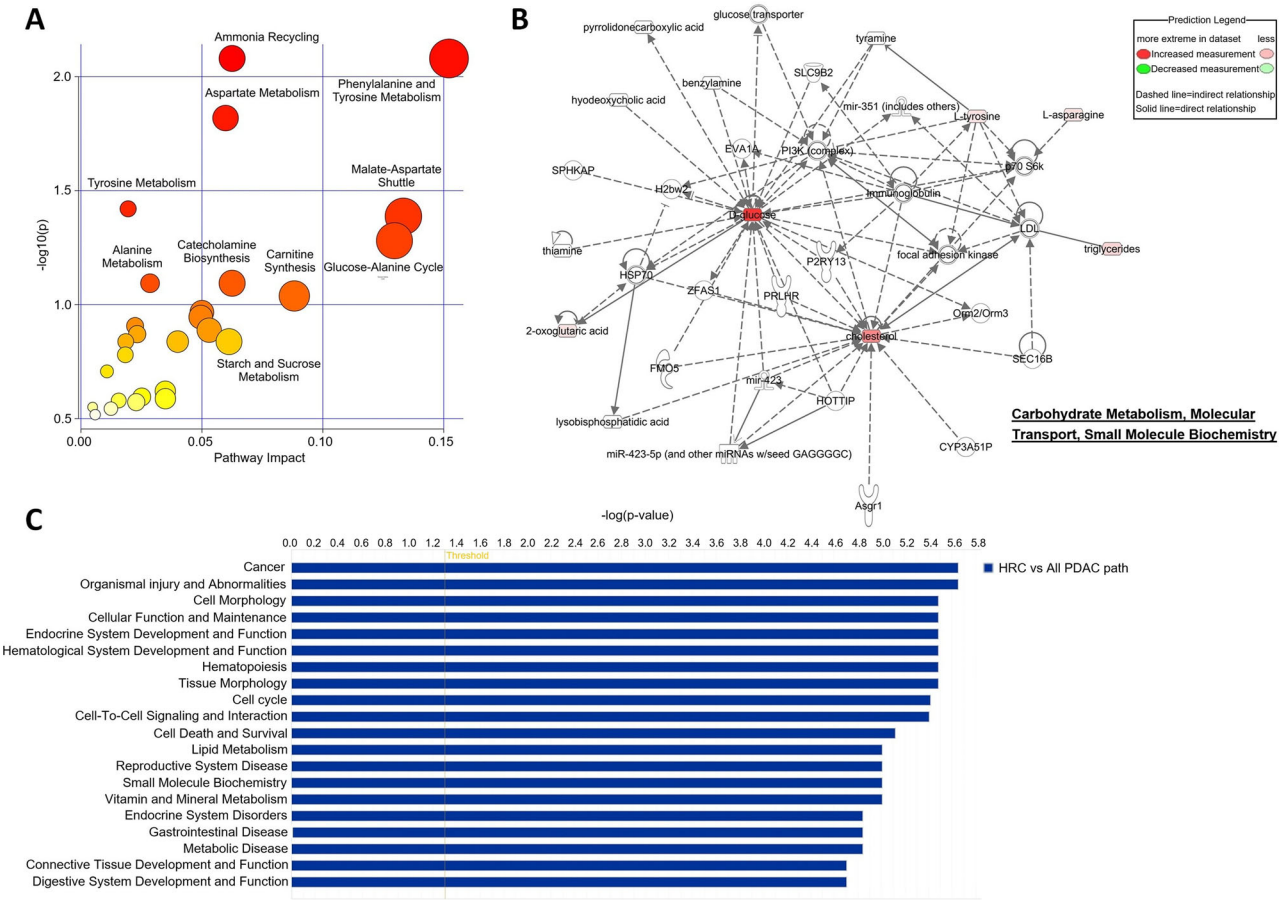

Fig. 5 | Pathway and network analysis of SHAP-identified serum metabolites in PDAC. A Metabolic Pathway Enrichment Map: Pathway analysis performed using MetaboAnalyst 6.0. The graph shows metabolic pathways significantly enriched by SHAP-ranked metabolites (e.g., 3-Aminoisobutanoic acid, Glucose, Taurine, TMAO). The y-axis represents the enrichment analysis results ($-\log_{10}(P)$, the raw p-value is calculated using Global Test), while the x-axis indicates pathway impact scores (summarize normalized topology measures using Relative-betweeness Centrality of the perturbed metabolites in each pathway). The bubble sizes are correlated with the x axis, and the color gradients refer to the y axis. **B** Metabolic Interaction Network: A network generated by Ingenuity Pathway Analysis (IPA) illustrating interactions within Carbohydrate Metabolism, Molecular Transport, and Small Molecule Biochemistry. Red-colored nodes represent molecules found to be increased in PDAC patients (lighter colors indicate a less elevated trend), while white nodes have no directional prediction. Solid lines represent direct interactions, and dashed lines represent inferred indirect interactions. Arrows indicate the directionality of molecular influence or regulation. **C** Disease and Biofunctions Summary: The top 20 disease and functional categories identified by IPA based on the SHAP-identified metabolites, presented by the $-\log_{10}$(p-value) of enrichment. *P*-value is calculating using Fisher's Exact Test, with $-\log_{10}$(p-value) > 1.3 set as significant differences threshold. This validates the biological relevance of the selected features by linking them to hallmark themes in cancer progression, organismal injury, and lipid metabolism.

microenvironments[35,36]. Complementarily, metabolomics emerges as a powerful and holistic approach, offering a snapshot of the dynamic metabolic equilibrium within the body, encompassing both physiological and pathological states[14]. As the ultimate downstream consequence of intricate genetic and proteomic interactions, the metabolome provides a rich source of information for early disease diagnosis and continuous monitoring, including the notoriously challenging landscape of PDAC[18,21,37–39].The comparative analysis of metabolic profiles between healthy individuals and those with disease states paves the way for personalized early diagnosis and longitudinal disease surveillance[40].

While prior efforts in early PDAC detection have largely focused on mass spectrometry-based analyses of targeted metabolites, lipids, or microbial byproducts[18,21,38,39], these approaches often face limitations hindering clinical translation. The focus on restricted biomarker sets may fail to capture the comprehensive metabolic dysregulation inherent in complex diseases like cancer, potentially compromising diagnostic accuracy across diverse populations. While full spectral analysis using techniques like PESI-MS offers a more comprehensive view[41], reproducibility and standardization remain key hurdles for widespread clinical adoption of mass spectrometry-based platforms.

The rapid, stable, and reproducible nature of NMR spectroscopy is crucial for the deployment of clinical biomarkers. Our NMR-based approach, which adheres to the rigorous, standardized protocol of the Bruker IVDr platform (demonstrating low CVs of < 1.4% for sample handling and < 0.5% for machine measurement), enabled comprehensive metabolic and lipid fingerprinting across our large cohorts (Supplemental Table 4, Supplemental Figs. 1 and 2). This standardization is essential for facilitating cross-center data integration and enhancing the robustness of our findings. By integrating these high-quality NMR metabolomic data with clinical/biochemical parameters (CA19-9, age) and the protein biomarker Activin A using the efficient TabPFN algorithm, we developed TabPFN-PanMETAI. The validation of TabPFN-PanMETAI across three distinct cohorts from two geographically separate countries (Taiwan and Lithuania) yielded consistent performance, providing strong evidence of its robustness and applicability across diverse patient populations for the rapid and accurate detection of early PDAC.

An intriguing finding was that a relatively small subset (~250) of NMR-derived metabolic features from NMR-M and NMR-M + L effectively distinguished HRC from PDAC within TabPFN. Assigning these signals revealed key metabolic alterations: dysregulated lipid

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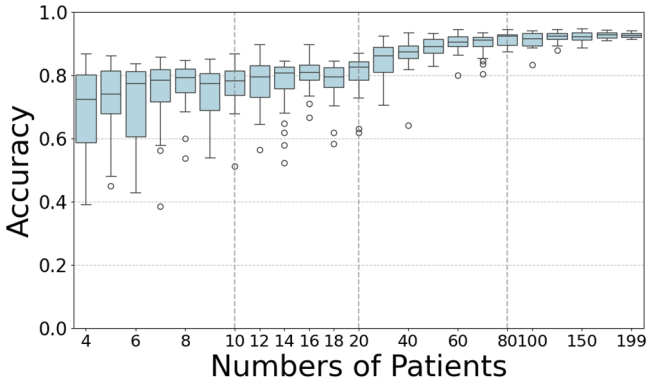

Fig. 6 | TabPFN-PanMETAI achieves high performance with limited training sample size. Accuracy as a function of patient cohort size ($n = 4$–199). The x-axis uses irregular intervals with unit increments for $n \leq 10$, increments of 2 for $10 < n \leq 20$, increments of 10 for $20 < n \leq 80$, and 100, 130, 150, 180, 199. Vertical dashed lines separate regions with different increments. Boxes represent the interquartile range (IQR), with horizontal lines in the boxes indicate the median, whisks for 1.5× IQR, and individual points for outliers. The corresponding AUC, sensitivity, and specificity are reported in Supplemental Figure 8.

metabolism (HDL decreased, VLDL increased), upregulated glycolysis (glucose increased, lactate increased), and perturbed amino acid metabolism (glutamine decreased, glutamate increased, alanine increased, aspartate increased). These findings align with previous reports documenting glutamine metabolism as a fuel source supporting cellular redox balance and cancer growth[26,42], and the utilization of the increased glycolysis product, lactic acid, by cancer-associated fibroblasts to promote immunosuppression in PDAC[43]. Furthermore, altered levels of lipoproteins, glycerophosphocholine (GPC), and glutamic acid have been associated with PDAC progression and identified as potential biomarkers[27,28,44]. The interplay between altered lipid metabolism (VLDL, HDL) and microenvironment (Epithelial–mesenchymal transition, cell invasion, and angiogenesis) can contribute to cancer development[45,46], a well-established promoter of tumorigenesis. This observation aligns with previous reports[28,47,48] demonstrating that ^{1}H NMR spectroscopy enables sensitively detect metabolic alterations in pancreatic cancer. It further underscores NMR's strength in capturing interconnected metabolic changes using small feature subsets, underscoring the interpretability and biological relevance of the selected metabolic features in TabPFN-PanMETAI. Crucially, we rigorously tested the model's robustness by performing multiple subgroup analyses on the blind test set (Cohort 2, Supplemental Table 5). After controlling for major potential confounders (age, BMI, and smoking status), the model maintained high performance with an AUC of 0.96-0.97 (Supplemental Tables 6 and 7). To exclude potential bias from jaundice (which may elevate CA19-9), we analyzed the non-jaundiced subset (Supplemental Table 8), finding that the lower sensitivity observed in this subgroup (0.63 vs. 0.78 in jaundiced patients) was consistent with the loss of the diagnostic boost provided by jaundice-associated CA19-9 elevation (Supplemental Table 9). Finally, evaluation across multiple sampling time points (spanning from 2005 to 2022; Supplemental Figure 9) demonstrated the high temporal stability of the predictions. Collectively, these multifaceted findings underscore TabPFN-PanMETAI's remarkable resilience to clinical heterogeneity and temporal drift, reinforcing its strong potential for reliable deployment in real-world clinical settings.

Beyond accuracy, TabPFN-PanMETAI's computational efficiency via TabPFN offers a significant advantage for clinical translation. Unlike models requiring large, multi-center datasets, TabPFN's low resource demands allow high-performing models to be trained on small, single-institution cohorts (50-70 samples; Fig. 6). The minimal GPU memory suggests development on standard hardware is feasible, enabling individual hospitals to build tailored diagnostic models, reducing data collection burdens, and offering a practical approach for precision oncology. Another core strength of TabPFN-PanMETAI lies in its adoption of the TabPFN algorithm. Built on an in-context learning approach and pre-trained on a vast number of synthetic datasets, TabPFN is uniquely capable of generating accurate predictions without requiring additional optimization or hyperparameter tuning. This feature makes it practical for heterogeneous datasets derived from different institutions. Furthermore, TabPFN demonstrates intrinsic robustness to missing values and outliers[49]. Its established capability to handle up to 10,000 samples and 500 features[50], often outperforming deep tabular models and AutoML systems like AutoGluon, reinforces its suitability and scalability for complex, multi-center clinical data analysis.

Despite these promising findings, some limitations remain. While TabPFN integrates technical interpretability tools like SHAP and LOCO, its Transformer-based architecture is inherently less transparent than traditional methods like linear regression. Furthermore, while SHAP-based annotations enabled robust pathway analyses (Fig. 5) that revealed distinct metabolic associations with pancreatic cancer, the underlying biological mechanisms remain speculative and require independent laboratory validation. Finally, we must address the technical limitations of ^{1}H NMR spectroscopy, particularly its difficulty in resolving overlapping signals in crowded spectral regions, which can constrain the confident identification of certain lipid species. To overcome this technical challenge and enhance diagnostic performance, future iterations of the TabPFN-PanMETAI model will integrate complementary mass spectroscopy-based lipidomic profiling. This addition aims to expand metabolite coverage, particularly for informative biomarkers like long-chain sphingomyelins, ceramides, and (lyso)phosphatidylcholines[19].

In conclusion, this study develops TabPFN-PanMETAI, a rapid, reliable, comprehensive, minimally invasive, and highly specific methodology representing a substantial advancement in early PDAC diagnostics. The high accuracy achieved by integrating metabolomic, clinical/biochemical parameters, and protein data offers significant potential for improved patient outcomes through early intervention and widespread applicability in screening and diagnosis, due to its non-invasive nature. However, large-scale, prospective, international studies in diverse high-risk individuals and very early-stage PDAC patients are crucial to definitively validate its clinical utility and facilitate broader implementation.

Methods

Study population

Our study analyzed a total of 1224 serum samples collected from two geographically distinct institutions: 902 from the National Taiwan University Hospital (NTUH) (January 2004–December 2022) and 322 from the Lithuanian University of Health Sciences (LSMU) (October 2015–June 2024). The study cohort was systematically partitioned to support the development and validation of machine learning. A total of 478 confirmed PDAC patients and 424 high-risk individuals (HRIs) with a family history of PDAC from Taiwan comprised the training and validation cohort, whereas 172 confirmed PDAC patients and 150 control subjects from Lithuania constituted the external validation cohort. All patients provided written informed consent prior to treatment, and diagnoses were pathologically confirmed. Samples were obtained under standardized fasting conditions and isolated from peripheral blood. This comprehensive cohort structure complies with all relevant ethical regulations, approved by both the NTUH Institutional Review Board (201301048RIND) and the Kaunas Regional Biomedical Research Ethics Committee (BE-2-31), was essential for developing a robust predictive model and validating its generalizability across diverse clinical settings (Participant demographics are detailed in Table 1 and Table 2).

Serum sample preparation
All serum samples were prepared from blood collected under fasting conditions using a serum-separating tube (BD Vacutainer Systems, Franklin Lakes, NJ, USA) following a standard venous blood sampling protocol. The samples were centrifuged at 3000 *g* for 10 min at 4 °C, then stored at −80 °C until use. Before ^{1}H NMR spectroscopic analysis, frozen serum samples were thawed on ice for 30 min and centrifuged for 10 min at 13000 *g* at 4 °C. Serum samples were prepared using 3 mm Ø, 5-long SampleJet NMR tubes (Brucker BioSpin GmbH, Rheinstetten, Germany). For standard sample preparation, 110 μL of serum was mixed with 110 μL phosphate buffer (75 mM Na_2HPO_4, 2 mM NaN_3, 4·6 mM TSP in D_2O), centrifuged for 10 s, and 200 μL was transferred to a 3 mm NMR tube for analyses.

^{1}H NMR acquisition
The ^{1}H NMR spectra were acquired using a 600 MHz Bruker Avance III HD spectrometer equipped with a 5 mm BBI probe and fitted with the Bruker SampleJet robot cooling system set to 4 °C. A standard one-dimensional (1D) pulse sequence with solvent presaturation experiment included both small molecule metabolites and lipoproteins (32 scans, zero-filled to 128 K refers to NMR-M + L spectrum), whereas a Carr−Purcell−Meiboom−Gill spin−echo experiment was used to identify small molecule metabolites (32 scans, zero-filled to 128 K refers to NMR-M spectrum). The data were processed in automation using a Bruker Topspin 3·6·2 and ICON NMR to achieve phasing and baseline correction. The TSP-d_4 provided a chemical shift reference (1H, δ 0·00). All spectra passed the Bruker in vitro diagnostics research (IVDr) validation criteria[51]—covering water suppression quality, line width (≤1.50 Hz for reference metabolite, Alanine), and shim—were included for downstream machine learning analysis.

The quantification of serum metabolites and lipoproteins were completed using the Bruker IVDr methods. A total of 39 metabolites and 112 lipoprotein parameters were generated using the IVDr Quantification in Plasma/Serum (B.I.Quant-PS) and Lipoprotein Subclass Analysis (B.I.LISA) methods[52].

Data processing of ^{1}H NMR spectra for ML
Two types of spectra (NMR-M & -M + L) were measured, one is metabolite-specific (NMR-M) and another, lipoprotein-containing (NMR-M + L). The NMR spectra were read using the nmrglue module[53]. A Fourier-transformed NMR spectrum contains a list of signal intensities corresponding to the ^{1}H chemical shifts. After acquiring the reads, each spectrum was normalized by its internal standard TSP-d_4 for handling bench differences. The normalized spectrum was further trimmed for removing regions that do not contain meaningful information: for NMR-M, the data with frequencies lower than or equal to 0.13 ppm, or higher than or equal to 11.13 ppm were removed; for NMR-M + L, lower than or equal to 0.11ppm or higher than or equal to 10.00 ppm were trimmed; for both NMR spectra, data between 4.64 ppm and 4.78 ppm was removed due to the water signal. After normalization and trimming, the resulting spectra were used as inputs for the following ML pipeline.

ML Feature Selection
Each signal intensity in the spectrum is considered a separate feature. As the trimmed NMR spectrum contains 71,074 (NMR-M) and 42,500 (NMR-M + L) features, we further reduce their number using feature selection in order to extract the most relevant information.

We rank features using different methods: NMR-M with f_classif, an F-test-based algorithm, and NMR-M + L with a linear SVM, both from the *scikit-learn* package (version 1·2·2). Feature selection is performed using SelectKBest for NMR-M and recursive feature elimination (RFE) for NMR-M + L, both from *scikit-learn*. The different methods chosen for NMR-M and NMR-M + L are motivated by differences in their spectral profiles, as NMR-M + L generally contains broader signals. In order to access the discriminating power, selected features are clustered using UMAP[54] [umap package (version 0.5.7)], and the Silhouette score [cuml package (version 24.10.00)] for clusters of healthy and diseased samples is calculated. With an iteration through different numbers of features picked, we searched for the features that yield the highest Silhouette score, and the corresponding features are used. With this process, we selected 1,563 features of NMR-M and 6,800 features of NMR-M + L for the multilayer SVM and AutoGluon models. For TabPFN, due to the constraints on feature numbers, we select 215 features of NMR-M and 255 features of NMR-M + L.

Missing data handling
All patients with missing or NA (Not Applicable) values in clinical/biochemical records are excluded from the dataset to provide a fair basis for benchmarking (Table S1) for all ML models. The precise numbers for each model are presented in ROC-AUC figures.

Model evaluation pipeline
The training dataset was used to train the model, and the Dev-Test was used for multilayer hyperparameters (HPs) tuning. Four indices were used to benchmark each model's performance: accuracy, specificity, sensitivity, and area under the receiver operating characteristic (ROC) curve (AUC). After HPs tuning upon Dev-Test, the tuned models (multilayer SVM), XGBoost from AutoGluon, and TabPFN were compared, and the champion model was selected for the final performance test on the blind test dataset.

Model architectures
AutoGluon provides detailed descriptions for each of its implemented models: https://auto.gluon.ai/stable/api/autogluon.tabular.models.html[23].

The description of TabPFN can be found in their original paper[22].

The customized architecture for multilayer SVM is arranged to handle the multimodal nature of our inputs. This architecture is composed of 3 layers: *encoder*, *fusion*, and *final decision* layers.

Encoder. For multimodal learning, an "encoder" was used to transform different data modalities into a representation to fuse later. We first scaled the input data with StandardScaler function [*scikit-learn* package (version 1.2.2)], followed by two independent SVMs derived from the two NMR spectra with their values of radial basis function (RBF) kernel as the "decision value." We extracted such a value from the SVM as the new representation of the original spectrum, and it was subsequently fused with clinical/biochemical parameters and protein data.

Fusion. In general, there are several methods for fusing modalities in multimodal learning. Here, we simply chose to concatenate the data. Following the encoder layer, the decision values were concatenated with clinical/biochemical parameters and protein data.

Final decision. The concatenated features were scaled again using StandardScaler, followed by another SVM for the final decisions.

Measurement of Activin A levels in serum
Serum levels of Activin A were quantified using commercially available sandwich enzyme-linked immunosorbent assay kits (DAC00B, R&D Systems, Inc.) based on the manufacturer's instructions. Assays were performed duplicate for both biomarkers. Activin A levels were measured after a twofold dilution of the serum samples, followed by absorbance measurement at 450 nm with the same spectrophotometer. Activin A data is presented as pg/mg, which was normalized to the total protein (mg/mL) in the serum sample.

Statistical analysis

The biochemical results are presented as the means and standard deviations. They were analyzed according to the statistical methods available in GraphPad Prism 10·0 (San Diego, CA). Most metrics in ML models are neither normally distributed (confirmed by Kolmogorov–Smirnov test) nor having an equal variance (confirmed by Levene test), we performed Wilcoxon Signed Rank test to tackle this data distribution. All statistical tests were performed using the stats module from the SciPy library (version 1.14.1). Significant differences were presented with P-values < 0·05.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The raw NMR spectral data from the three participant cohorts supporting the findings of this study have been deposited in the Zenodo repository under the https://doi.org/10.5281/zenodo.18080179. Due to the potential risk of re-identification associated with high-dimensional metabolic profiles integrated with specific clinical characteristics, the dataset is available under controlled access to protect participant privacy in accordance with institutional ethical guidelines. Access to the de-identified dataset will be granted to bona fide researchers for non-commercial academic research purposes designed to reproduce the findings or validate the models presented in this study. Researchers wishing to access the data must submit a formal request to the corresponding authors (Chao-Ping Hsu /cherri@as.edu.tw; Yu-Ting Chang /yutingchang@ntu.edu.tw; Chun-Mei Hu /CMHU1220@as.edu.tw), providing their institutional affiliation and a brief statement of the intended scientific use. Requests from anonymous or unverified sources will not be granted. Access is conditional upon institutional review and the signing of a Data Sharing Agreement (DSA) that explicitly prohibits commercial usage, modification, and redistribution of the data. The authors commit to reviewing and responding to all completed access requests within approximately seven working days. Patient-identifiable information is strictly withheld to protect participant privacy.

Code availability

The source code and trained model weights supporting the findings of this study are protected by patents [MULTIMODAL MACHINE LEARNING SYSTEM AND METHOD; PCT/US25/51228; 114139299]. To safeguard this intellectual property while facilitating academic reproducibility, the TabPFN-PanMETAI implementation framework is available under a managed access protocol via Zenodo with the https://doi.org/10.5281/zenodo.18365111. Access will be granted to academic researchers for non-commercial research purposes upon request through the repository platform. All the access is granted based on PolyForm Noncommercial License 1.0.0 license for academic use only, which explicitly prohibits commercial use and modification. Commercial entities interested in the model or clinical deployment are encouraged to contact the corresponding authors directly. The code used to implement the comparator AutoGluon models has been deposited in a publicly accessible GitHub (https://github.com/PanMETAI/AutoGluon_PDAC) and archived on Zenodo (https://doi.org/10.5281/zenodo.17701588), protected by PolyForm Noncommercial License 1.0.0 license. All scripts required for installation steps, environmental requirements, and execution examples are provided to facilitate reproducibility. The TabPFN-PanMETAI approach utilizes the openly available foundation model TabPFN, which can be accessed directly at https://github.com/PriorLabs/TabPFN.

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Author contributions

D.N.W. contributed to the data curation, investigation, methodology, formal analysis, data interpretation, and writing-original draft. J.J. contributed to the methodology, formal analysis, and writing-original draft. E.F. contributed to the methodology. M.F.H., J.C.Y., and K.S. contributed to the formal analysis. M.C.C., J.K., J.S., R.S., E.M., and G.G. provided clinical serum samples. C.P.H., Y.T.C., and C.M.H. played key roles in conceptualizing the study, providing supervision throughout the research process, and reviewing and editing the manuscript. All authors confirmed the accuracy of the data and analysis, participated in revisions, approved the final version, and agreed to its publication. They also had full access to all study data and shared final responsibility for submitting the manuscript for publication.

Competing interests

The authors [Dan-Ni Wu, Chao-Ping Hsu, Yu-Ting Chang, Chun-Mei Hu] are listed as inventors on a filed patent application [MULTIMODAL MACHINE LEARNING SYSTEM AND METHOD; PCT/US25/51228;

114139299] related to PanMETAI models and their application in disease detection. No other competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Chao-Ping Hsu, Yu-Ting Chang or Chun-Mei Hu.

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